LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of shaft axis, axis rotation, and plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit of RPM
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of shaft axis, axis rotation, and plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit of RPM.
5. Specific Objectives	12.1.2.1 Trainee will be able to understand various terminology associated with helicopters.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of rotor terminology diagrams; Instructor supervision on diagram labeling
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the rotor system?
(b) Can you list three components of the rotor assembly?
(c) Why is understanding rotor terminology important for maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Shaft axis definition • Axis rotation direction • Plane of rotation • Tip path plane characteristics • Blade pitch definition	L & Ds	PPT, IB	15 min
(b)	• Rotor thrust generation • Relative air flow concepts • Coning angle explanation • Limit of RPM parameters • Interaction of thrust and airflow	L & Ds	PPT, PDF	20 min
(c)	• Summary of rotor terminology • Application of terms in context • Clarification of complex concepts • Review of shaft and pitch • Final Q&A session	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Define the tip path plane and its relationship to the rotor hub.
(b) Explain how blade pitch affects rotor thrust generation.
(c) What factors determine the limit of RPM for a specific rotor system?

12. Home Assignment and Follow Up	Prepare a brief note on the components to be covered in LP 03 (rotor mast, swash plate, torque link) and how they relate to the rotor terminology defined in this lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Components of rotor mast, control system, swash plate, torque link, drag damper, droop restrainer, clutch mechanism, rotor brake, feathering sleeve etc.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min